

AKRON-CANTON, OH, USA

WMO: 725210

Lat: 40.918N

Lon: 81.444W

Elev: 370

StdP: 96.96

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-13.7	-20.8	0.6	-15.8	-18.2	0.8	-12.7	11.9	0.5	11.0	-0.8	4.1	250	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.5	22.7	30.2	22.0	28.9	21.2	24.1	29.4	23.3	28.0	22.5	26.9	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.0	27.0	21.7	17.1	25.8	21.0	16.3	25.1	74.4	29.5	70.9	28.2	67.7	27.0	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.4	8.9	8.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-19.7	33.3	3.7	1.6	-22.4	34.5	-24.5	35.4	-26.6	36.3	-29.3	37.4
			-20.1	26.0	3.6	1.5	-22.7	27.1	-24.8	28.0	-26.8	28.8	-29.5	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-3.0	-1.5	3.2	10.0	15.8	20.6	22.6	21.9	18.2	11.7	5.4	0.1		
	DBStd	10.34	6.76	6.32	6.27	5.30	4.75	3.66	2.90	2.86	4.00	4.85	5.23	5.65		
	HDD10.0	1532	406	325	227	64	7	0	0	0	1	37	157	309		
	HDD18.3	3291	662	556	468	253	109	21	2	5	50	212	389	564		
	CDD10.0	1710	2	3	17	65	186	318	391	368	247	91	18	3		
	CDD18.3	427	0	0	1	3	30	89	136	115	47	7	0	0		
	CDH23.3	3244	0	0	6	38	247	669	1073	826	344	41	1	0		
	CDH26.7	906	0	0	0	3	50	191	335	228	94	6	0	0		
(14) Wind	WSAvg	4.0	4.8	4.6	4.5	4.5	3.8	3.4	3.1	3.0	3.2	3.8	4.3	4.5		
(15) Precipitation	PrecAvg	989	63	58	81	88	100	97	102	89	87	70	77	73		
	PrecMax	1669	156	145	224	164	207	236	319	208	229	180	238	171		
	PrecMin	604	18	8	32	23	27	9	17	12	15	12	16	22		
	PrecStd	186	31	28	33	36	46	53	62	48	46	35	38	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	18.2	23.3	26.9	30.1	32.2	33.3	32.4	32.1	27.3	20.7	16.4		
		MCWB	12.8	13.3	15.1	17.3	20.7	22.7	24.4	22.9	21.9	20.2	14.8	13.3		
	2%	DB	13.3	13.9	19.3	24.2	28.0	30.6	31.6	30.8	29.4	24.1	18.0	13.7		
		MCWB	11.5	10.7	12.9	15.7	19.6	22.0	23.3	22.1	21.3	18.2	13.7	11.4		
	5%	DB	10.2	10.6	16.2	21.6	26.3	29.0	30.0	29.2	27.0	21.9	15.8	11.2		
		MCWB	8.2	7.7	11.6	14.3	18.8	21.3	22.3	21.6	20.2	16.8	12.1	9.1		
	10%	DB	6.7	7.4	13.3	19.1	24.1	27.3	28.6	27.7	25.0	19.7	13.6	8.4		
		MCWB	4.6	5.0	9.4	12.8	17.8	20.3	21.3	20.9	19.1	15.5	10.5	6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	14.2	15.8	18.9	22.4	24.5	25.9	25.0	23.9	21.5	16.7	14.5		
		MCDB	15.2	17.7	21.7	24.5	28.0	30.1	31.0	29.9	29.7	25.4	19.6	15.8		
	2%	WB	11.9	11.1	14.0	17.0	21.0	23.2	24.5	23.7	22.4	19.5	14.6	11.9		
		MCDB	13.4	13.1	17.9	22.1	26.2	28.5	29.6	28.5	27.1	22.6	17.0	13.4		
	5%	WB	8.4	7.9	12.1	15.5	19.9	22.3	23.4	22.8	21.3	17.6	12.7	9.2		
		MCDB	9.9	10.3	15.6	20.0	24.7	27.2	28.2	27.0	25.5	20.7	15.1	10.7		
	10%	WB	4.8	5.1	9.8	13.9	18.6	21.4	22.5	21.9	20.2	15.8	10.8	6.5		
		MCDB	6.5	7.2	13.2	17.9	23.0	25.7	26.8	25.7	23.9	19.1	13.3	8.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	9.5	11.1	10.8	10.3	10.3	10.1	10.3	9.6	8.2	6.8		
		MCDBR	10.2	11.9	13.4	13.9	12.6	11.7	11.3	11.3	12.0	11.8	11.3	9.8		
	5% WB	MCWBR	8.9	9.6	8.7	8.3	6.5	5.5	5.0	4.9	5.5	6.8	8.2	8.3		
		MCWBR	9.8	11.2	12.0	12.1	11.1	10.4	10.1	9.7	10.1	10.2	10.2	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.324	0.352	0.385	0.423	0.446	0.447	0.436	0.381	0.358	0.315	0.301		
		taud	2.434	2.347	2.351	2.291	2.209	2.182	2.196	2.218	2.347	2.405	2.501	2.467		
	Ebn at Noon		860	884	897	887	857	835	829	828	855	834	836	827		
		Edh at Noon	79	100	112	127	141	146	142	135	111	94	74	71		
(44) All-Sky Solar Radiation	RadAvg		1.44	2.32	3.29	4.49	5.36	5.87	5.86	5.29	4.26	2.79	1.81	1.21		
	RadStd		0.13	0.30	0.33	0.43	0.45	0.45	0.41	0.27	0.39	0.25	0.27	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.76	N/A	N/A	+0.72	N/A	N/A	N/A	-181	+179	+103
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+63

Nomenclature: See separate page